ASSIGNMENT 1 — Q9

Random Math Quiz Generator

JavaScript solution

```
// Q9: Random Math Quiz Generator
"use strict";

const firstNumber = Math.floor(Math.random() * 20) + 1;
const secondNumber = Math.floor(Math.random() * 20) + 1;
const operators = ["+", "-", "*", "/"];
const operator = operators[Math.floor(Math.random() * operators.length)];

let correctAnswer;
switch (operator) {
    case "+":
        correctAnswer = firstNumber + secondNumber;
        break;
    case "-":
        correctAnswer = firstNumber - secondNumber;
        break;
    case "*":
        correctAnswer = firstNumber * secondNumber;
        break;
    case "/":
        correctAnswer = firstNumber / secondNumber;
        break;
    default:
        throw new Error("Unsupported operator");
}

if (operator === "/") {
    correctAnswer = Number(correctAnswer.toFixed(2));
}

console.log(`Question: ${firstNumber} ${operator} ${secondNumber} = ?`);
console.log(`Correct answer: ${operator === "/" ? correctAnswer.toFixed(2) : correctAnswer}`);
```

Logic: Randomly chooses two numbers and an operator, evaluates it with switch, and formats division to two decimals.

How to test: Save this code as a .js file and run it with Node.js using `node filename.js`, or paste it into the browser console.